

The empirical recurrence for OEIS A230945: recovering a conjecture that is not written down

Adrian Perez Fontelles
Independent researcher

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Abstract

OEIS A230945 records “Empirical recurrence of order 70”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 217$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 70, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 70 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A230945 is “Number of $(n+3) \times (4+3)$ 0..3 white square subarrays $x(i,j)$ with each element diagonally or antidiagonally next to at least one element with value $(x(i,j)+1) \bmod 4$, and upper left element zero.”. It has offset 1 and begins

314, 1994, 85196, 606224, 22328204, 161835904, 6209914254, 44707108238, ...

The entry states:

Empirical recurrence of order 70 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 06:08:05, revision 7) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 217$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 217$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 434$ exact terms of the model returns order 70 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{70} c_i a(n-i),$$

c_1	0	c_{19}	0	c_{37}	0	c_{55}	
c_2	228	c_{20}	2642403779905870	c_{38}	2917379659294469072447	c_{56}	−253870953
c_3	0	c_{21}	0	c_{39}	0	c_{57}	
c_4	15195	c_{22}	−107393060687303994	c_{40}	121931685971456506340899	c_{58}	1890978907
c_5	0	c_{23}	0	c_{41}	0	c_{59}	
c_6	−470965	c_{24}	1110541214823141222	c_{42}	−371771072105454726920288	c_{60}	−101243239
c_7	0	c_{25}	0	c_{43}	0	c_{61}	
c_8	−57209721	c_{26}	−5429362449768976624	c_{44}	559594933927764117661291	c_{62}	−112005677
c_9	0	c_{27}	0	c_{45}	0	c_{63}	
c_{10}	−86482160	c_{28}	3491676509271492542	c_{46}	−979087846024262868843129	c_{64}	1627971939
c_{11}	0	c_{29}	0	c_{47}	0	c_{65}	
c_{12}	58067909911	c_{30}	120279040553959802238	c_{48}	3824924461187357095585390	c_{66}	−85154344
c_{13}	0	c_{31}	0	c_{49}	0	c_{67}	
c_{14}	129138938491	c_{32}	−707019741140065896977	c_{50}	−3523763854072349017327224	c_{68}	−49123751
c_{15}	0	c_{33}	0	c_{51}	0	c_{69}	
c_{16}	−35888636878668	c_{34}	2453968786421516659780	c_{52}	−15280573148652030513646032	c_{70}	89783896
c_{17}	0	c_{35}	0	c_{53}	0		
c_{18}	437014504261008	c_{36}	−9254367374777255219481	c_{54}	−18434115456325236148071120		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 70, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $70 + 4$ terms removed returns order 70 as well, so $\deg q_{\min} = 70 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 15 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $70 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 70 that this sequence satisfies — and that family turns out to have exactly one member.

References

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